

Frederick (Tzujie) Huang

Curriculum Vitae

University of Texas Health Science Center at San Antonio
Dept. of Comprehensive Dentistry
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RESEARCH INTERESTS

Modeling, analysis, and design of functional materials using machine learning for structure-property-processing prediction. Broad topics of interest include machine learning for materials, materials characterization, informatics, and inverse design. Current work is in characterizing biological and composite materials, curating proprietary data for developing prediction models, and developing materials property data miners.

EDUCATION

University of North Carolina at Chapel Hill
B. S., with Distinction, in Biomedical Engineering

Chapel Hill, NC
2025

North Carolina State University
B. S., magna cum laude, in Biomedical Engineering
Joint program between College of Engineering and UNC School of Medicine

Raleigh, NC
2025

RESEARCH EXPERIENCE

University of Texas Health Science Center at San Antonio

San Antonio, TX

Dept. of Comprehensive Dentistry Research Associate (with Kyumin Whang, PhD)

October 2025–present

- Validated preliminary machine learning (ML) model predictions for polymer composite formulations through ISO 4049 experimentation.
- Curated proprietary formulations as training data and engineered features for improving ML model performance. Built local LLM-assisted materials property data scraper backend for dental literature.
- Designed and built custom laboratory instruments to increase composite sample production, reducing discrepancies in degree of conversion.
- Developed an antifouling, remineralizing polymeric nanocomposite, coupling a quaternary ammonium silane functionalized matrix with phosphate mineral fillers for secondary caries prevention.
- Instructed operation and safety on lab tests, most commonly with Instron, Konica Minolta, and ThermoFisher instruments. Wrote data cleaning, visualization, and analysis pipelines for acquired mechanical testing data.

University of North Carolina at Chapel Hill

Chapel Hill, NC

Joint Dept. of Biomedical Engineering Research Assistant (with Jeffrey Macdonald, PhD)

August 2024–May 2025

- Contributed to study aimed at developing: 1) pentaomic extraction method and 2) a lipidomic integrity number (LIN), a sample quality metric analogous to established integrity measures (PIN, DIN, RIN/DV200) used in transcriptomic and genomic analysis, standardizing quality assurance in future bioinformatics studies.
- Subproject (*scholarship awarded*) analyzed LC-MALDI-TOF lipidomic datasets from *R. norvegicus* brain and *E. fuscus* liver specimens to characterize the differential effects of sample preparation conditions (drying and dissolution/mixing) on lipid degradation.
- Identified candidate lipid classes, primarily phosphatidylcholine (PC) and phosphatidylethanolamine (PE), exhibiting greatest sensitivity to sample preparation-induced chemical degradation, annotating data through LipidSearch 5.
- Developed data visualizations to evaluate lipid abundance across preparation conditions, characterizing lipid classes with statistically significant fold-changes as candidates for LIN feature weighting.
- Contributed to rationale for LIN feature selection. Determined preparation-sensitive lipids appropriate in a computational sample quality metric, with goal of establishing SOPs for data acquisition.

University of North Carolina at Chapel Hill

Chapel Hill, NC

Eshelman School of Pharmacy Research Assistant (with ZhenZhen Chen, PhD)

December 2023–August 2024

- Synthesized liposomal carriers and metal nanoparticles for therapeutic and theranostic applications, including formulations for incorporating DMXAA; developed procedural proficiency in liposome preparation and colloidal characterization with zeta potentials, SEM, and image analysis. Project ended at 8 months following departure of the supervising postdoc.
- Maintained EH&S compliance protocols for handling cytotoxic compounds and nanomaterials.

PROFESSIONAL EXPERIENCE

Academic Startup

Product Development Intern

San Antonio, TX

June 2026–Present

- Owned early-stage IP patentability, state-of-the-art, and patent portfolio overview reports for pre-seed MedTech startup supported through UT Health San Antonio Office of Innovation and Strategic Partnerships to develop hardware for gastroenterology and emergency medicine applications.

Pulpdent

Engineering Intern

Watertown, MA

May 2026–Present

- Conducted ISO 4049 compliant testing on experimental composite formulations, performed statistical analysis to identify property discrepancies to ensure production consistency and process refinement.
- Owned materials characterization and statistical analysis on experimental composite resin batches, combining mechanical/surface testing with batch-variability assessment to support formulation development.

SELECTED PROJECTS

Automated Literature Mining Pipeline | *Data Extraction, Image Segmentation and Digitization* April 2026–present

- Integrated local and cloud-based LLM backend for figure-caption extraction into an open-source scientific figure analysis framework, enabling automated reading of figure captions with text/tabular context from born-digital PDFs.
- Utilized OpenCV for image segmentation of multipanel figures using whitespace projection-profile analysis with a contour-detection fallback to separate complex scientific images.
- Ported PaddleOCR into workflow to digitize graphical figures and capture time dependent features for ML training.
- Constructed validation and quality-assurance tests to quantify panel-segmentation accuracy, caption coverage, and caption-to-figure correspondence with automated integrity checks on the exported feature tables, establishing a measurable baseline prior to downstream extraction.
- Adapted an open-source autonomous-materials-discovery codebase (Argonne Polybot, JACS 2025) to dental materials science domain.

Antifouling, Remineralizing Dental Composites | *Mechanical Testing, Data Analysis, GLP* December 2025–present

- Prepared experimental composites formulated to address secondary decay with quaternary ammonium silane (QAS) functionalized polymeric composites.
- Acquired and cleaned material property data for 80+ unique samples.
- Drafted all source code required for exploratory data analysis, visualization, and interpretation among 12+ experiments.
- Implemented statistical packages to elucidate key insights behind mechanical behavior affected by experimental nanocomposite formulations with QAS functionalized matrices.

Orthotic Brace for UNC Physical Medicine and Rehabilitation | *Rapid Prototyping* August 2024–May 2025

- Conducted shadowing and clinical interviews to develop a chest brace for adolescent pectus carinatum patients, implementing on-the-fly calibration and real-time feedback following ISO 14971 and 13485 standards.
- Altered patient Haller index by an average of 0.2 with a projected 30% decrease in standard treatment time.

Predicting Landing Success of Reusable Rocket Boosters | *sklearn, Data Mining/Cleaning* January 2026–March 2026

- Scraped all launch data using SpaceX's Falcon 9 boosters since 2010 from SpaceX's REST API and Wikipedia to construct a traditional ML model predicting Stage 1 landing success.
- Scored prediction performance across varying regression and classification models, tuning parameters through GridSearchCV and yielding up to 88% accuracy evaluated via confusion matrices.

Simulating a *C. geographus* Inspired Insulin Analog | *PyMOL, Cluspro, Chimera* August 2024–December 2024

- Investigated limitations in commercial insulin products and INSR/IGF1 receptor binding ability of cone snail venom proteins, engineering rapid pharmacokinetic properties into a thermally stable, single-chain insulin analog.
- Computationally predicted 12+ energy minimized candidate molecules and conducted *in silico* protein-protein docking, benchmarking designs against predicted structures from Google DeepMind's AlphaFold.
- Combed NIH's FASTA database to identify candidate bacteria-sourced homologs for potential scale-up production.

TEACHING EXPERIENCE

Course Name/Number	Enrollment	Type	Year
BMME209, Materials Science of Biomaterials	37	Ugrad	Spring 2025
String Performance and Music Theory	2	Secondary School	2022-2025

PRESENTATIONS

- S. Mathur, K. S. Paye, **F. Huang**, M. N. Cole, M. J. Gilbert, M. K. Hederick. (2025). *The Brace Check—Data Collec-*

tion and Monitoring for Pectus Carinatum Patients, Lampe Joint Department of Biomedical Engineering Senior Design Symposium, Raleigh NC.

- **F. Huang**, C. Arciniega, B. Ehrmann, J. Macdonald. (2025). *Quality Management in Lipidomic Extractions to Develop a Lipidomic Integrity Number*, UNC-Chapel Hill Undergraduate Research Symposium, Chapel Hill NC.

SELECTED HONORS AND AWARDS

Abrams Scholar, <i>College of Engineering, North Carolina State University</i>	2025
IPEP Distinguished Scholar, <i>University of North Carolina School of Medicine</i>	2025
Dean's List, <i>University of North Carolina at Chapel Hill</i>	2021-2025

SKILLS AND TECHNICAL COMPETENCIES

- **Tools:** Python, MATLAB
- **Packages:** Numpy, pandas, sklearn, Matplotlib, Seaborn, SciPy.
- **Software:** PyMOL, ImageJ, Onshape, Fusion360, KiCAD, Chimera, STARE, Office Suite.
- **Characterization:** Contact Angle, Indentation, Mechanical Load Frames, FTIR, SEM, DSC, DMA.